Power Domination in Graphs

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Second Reader: Professor Mauro

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Introduction

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One method of monitoring these variables is to place sensors called phase measurement units (PMUs) at selected locations in the system.

Because of the high cost of a PMU, it is desirable to minimize the number of PMUs while maintaining the ability to monitor (observe) the entire system.

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- $V(G)$ includes possible locations where PMUs can be located
- $E(G)$ denotes the set of wires carrying electricity
- $S \subseteq V(G)$ is said to be a power dominating set (PDS) if:
  - Every vertex and edge in G are observed by S.
  - Observation is in accordance with the rules based on Ohm's Law and Kirchhoff's current law.

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3. Any edge connecting two observed vertices is observed.
4. If a vertex is of degree $k > 1$, and $k - 1$ of these incident edges are observed, then all k incident edges are observed.

Definitions and Terminology

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- The *closed neighborhood* of $v \in V(G)$ is $N_G[v] = N_G(v) \cup \{v\}$.
- The *neighborhood* of a vertex set S, $N_G[S] = \bigcup_{s \in S} N_G[s]$.

Power Domination Algorithm

Algorithm [?]:

Let $S \subseteq V(G)$ be the set of vertices where the PMUs are placed.

1. (Domination)
   Set $M(S) \leftarrow S \cup N(S)$

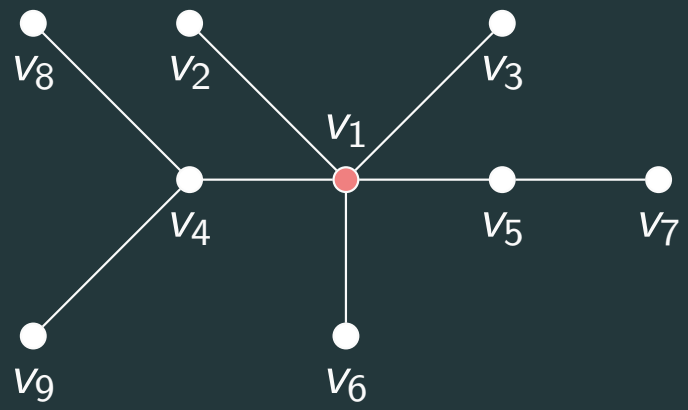


Figure 1: $S = \{v_1\}$

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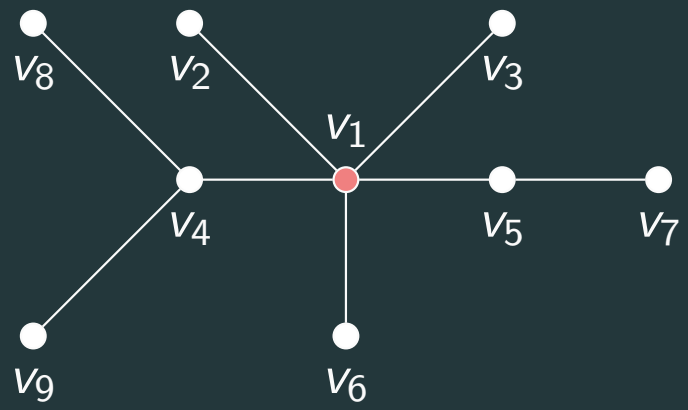


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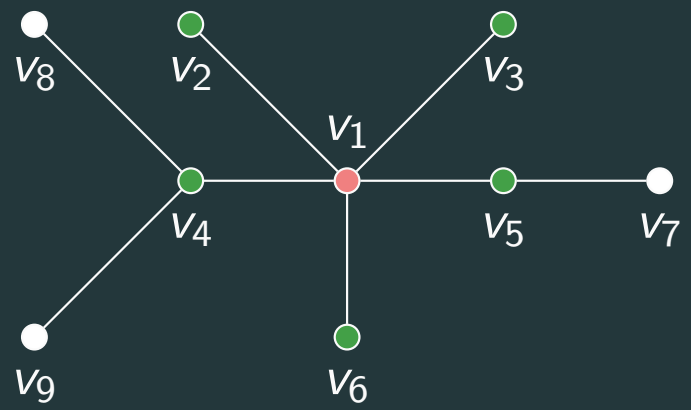


Figure 2: Domination Step

Power Domination Algorithm

Algorithm [?]:

2. (Propagation)
   As long as there exists $v \in M(S)$ such that
   $N(v) \cap (V(G) - M(S)) = \{w\}$, set $M(S) \leftarrow M(S) \cup \{w\}$.

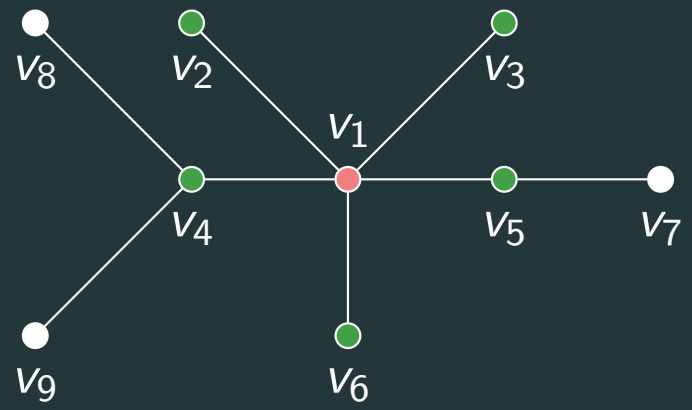


Figure 3: Observed vertices in previous time step: $N[v_1]$

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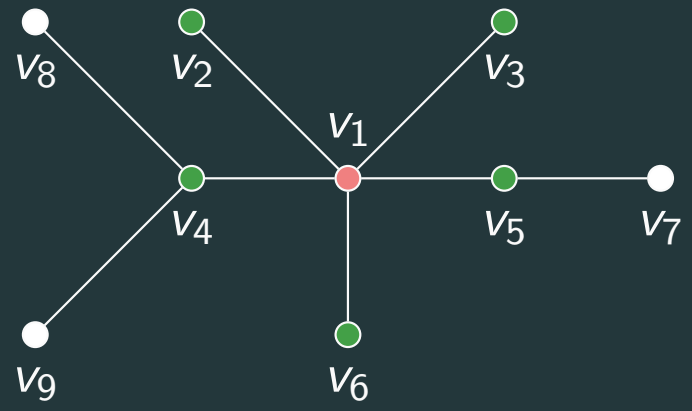


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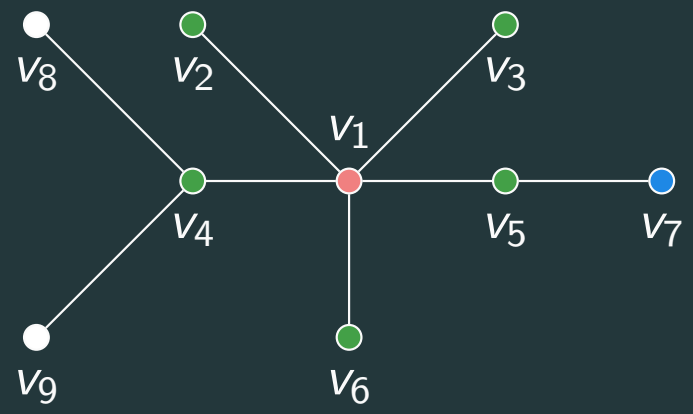


Figure 4: Propagation Step

Example ($S = v_1, v_{12}$)

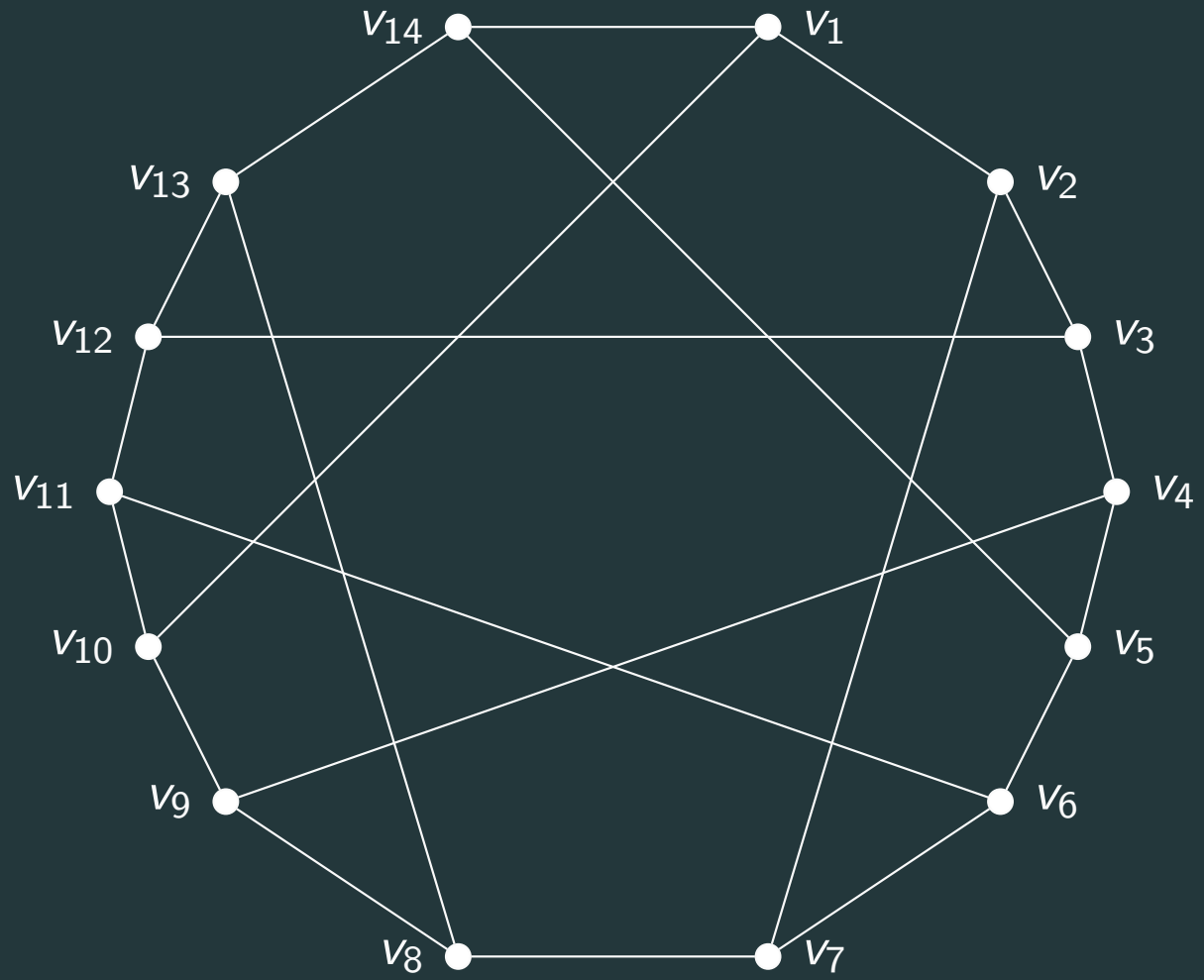


Figure 5: Haewood Graph

Power Domination for Haewood Graph

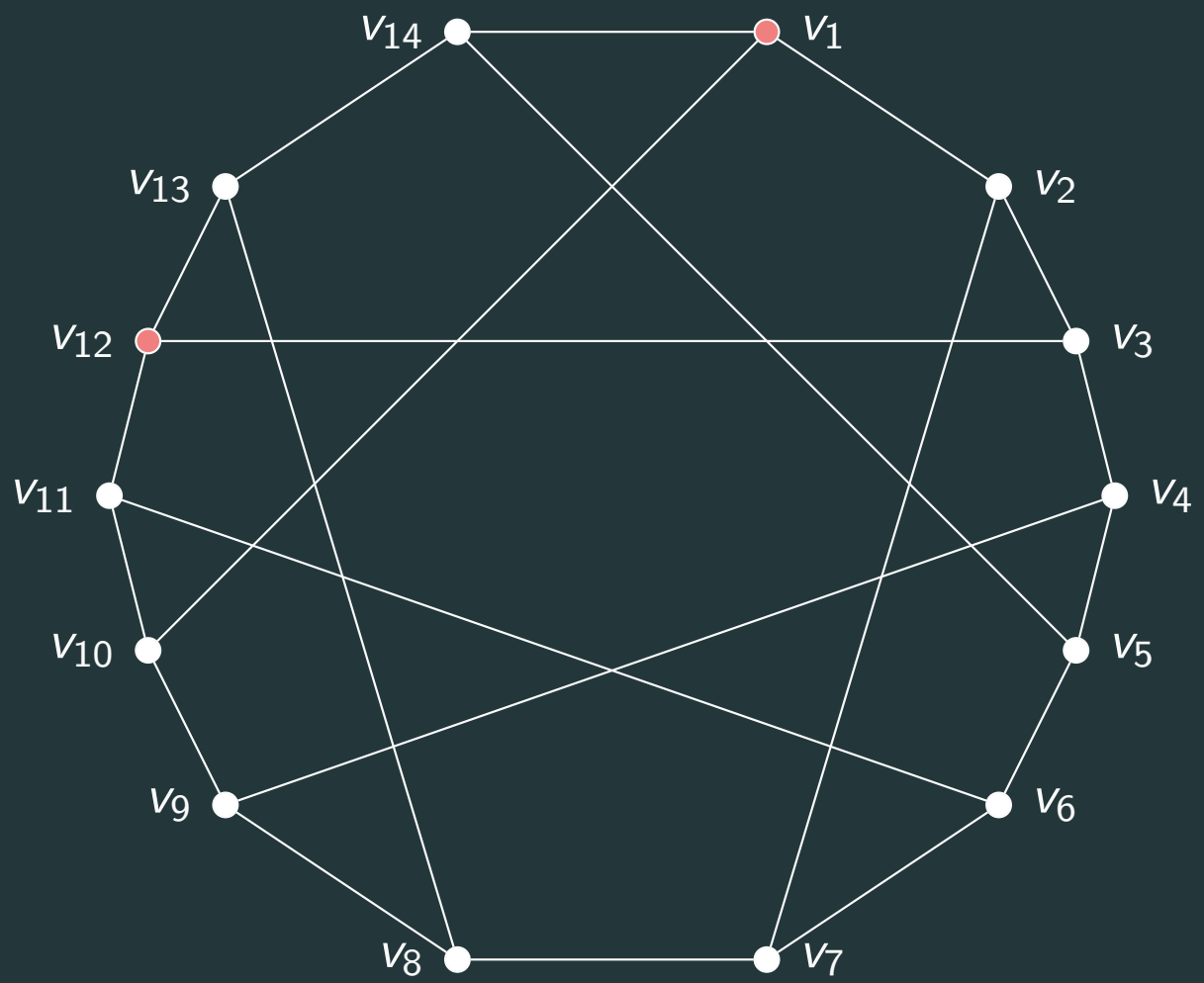


Figure 6: Haewood Graph

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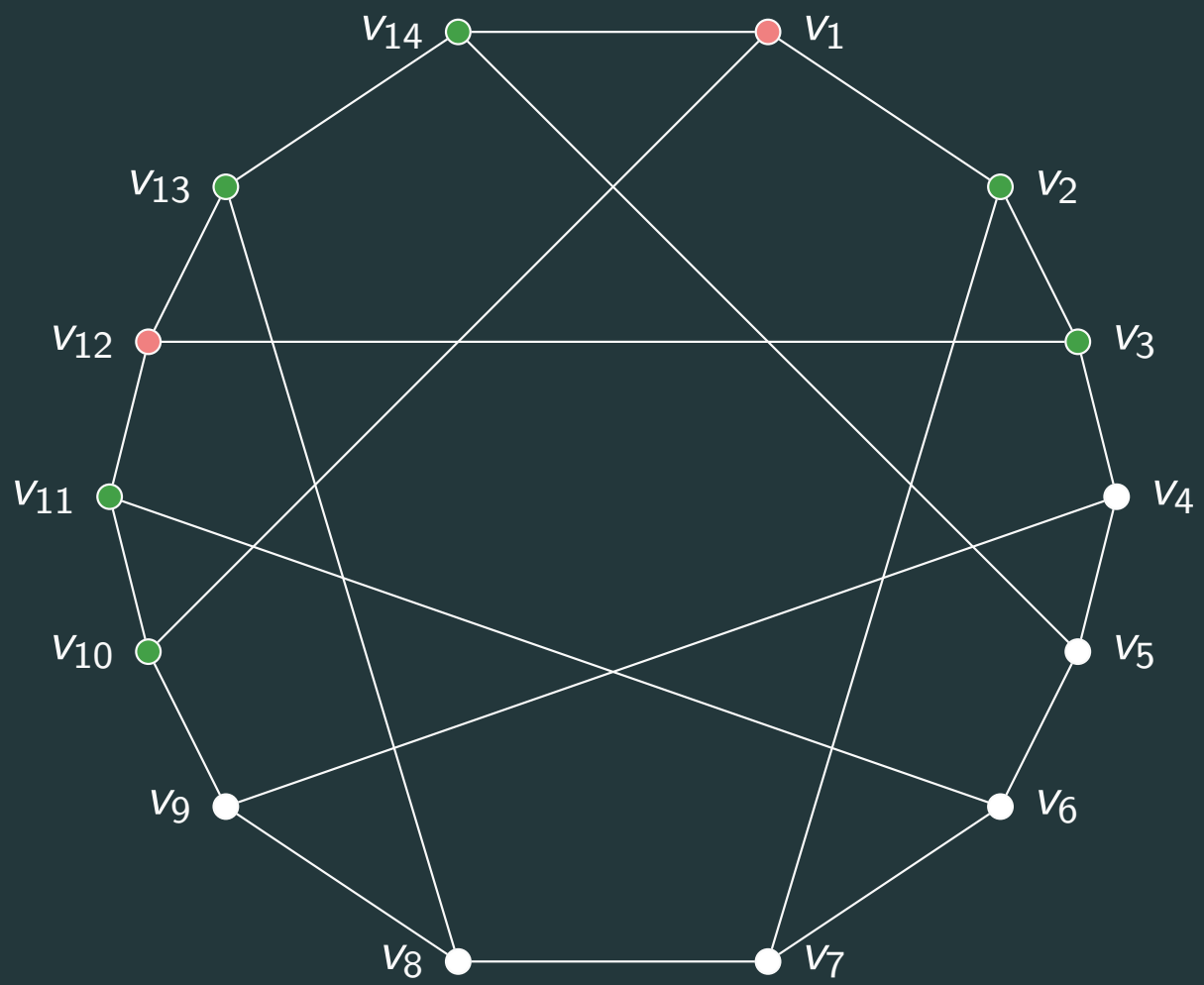


Figure 7: Haewood Graph

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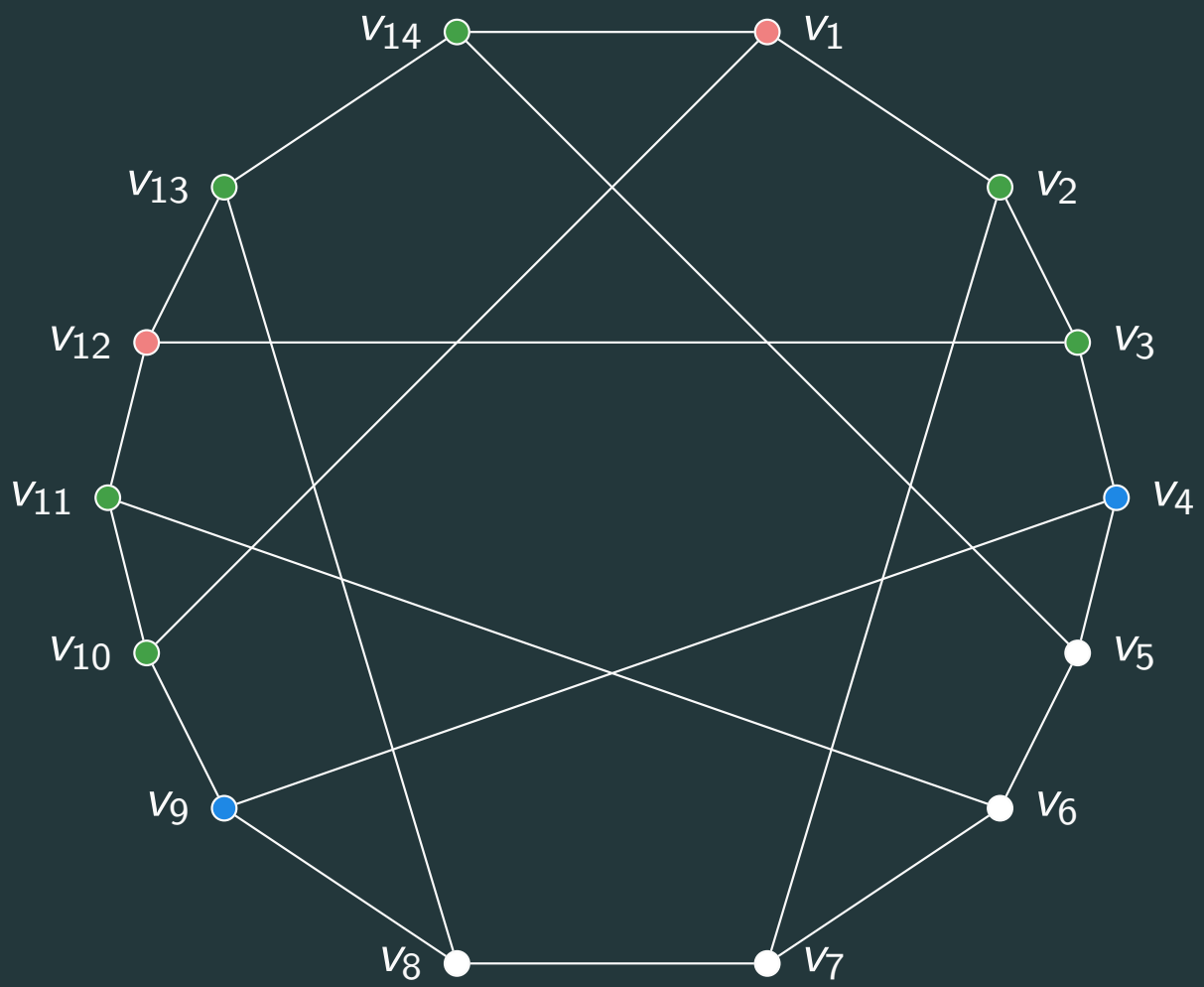


Figure 8: Haewood Graph

Power Domination for Haewood Graph

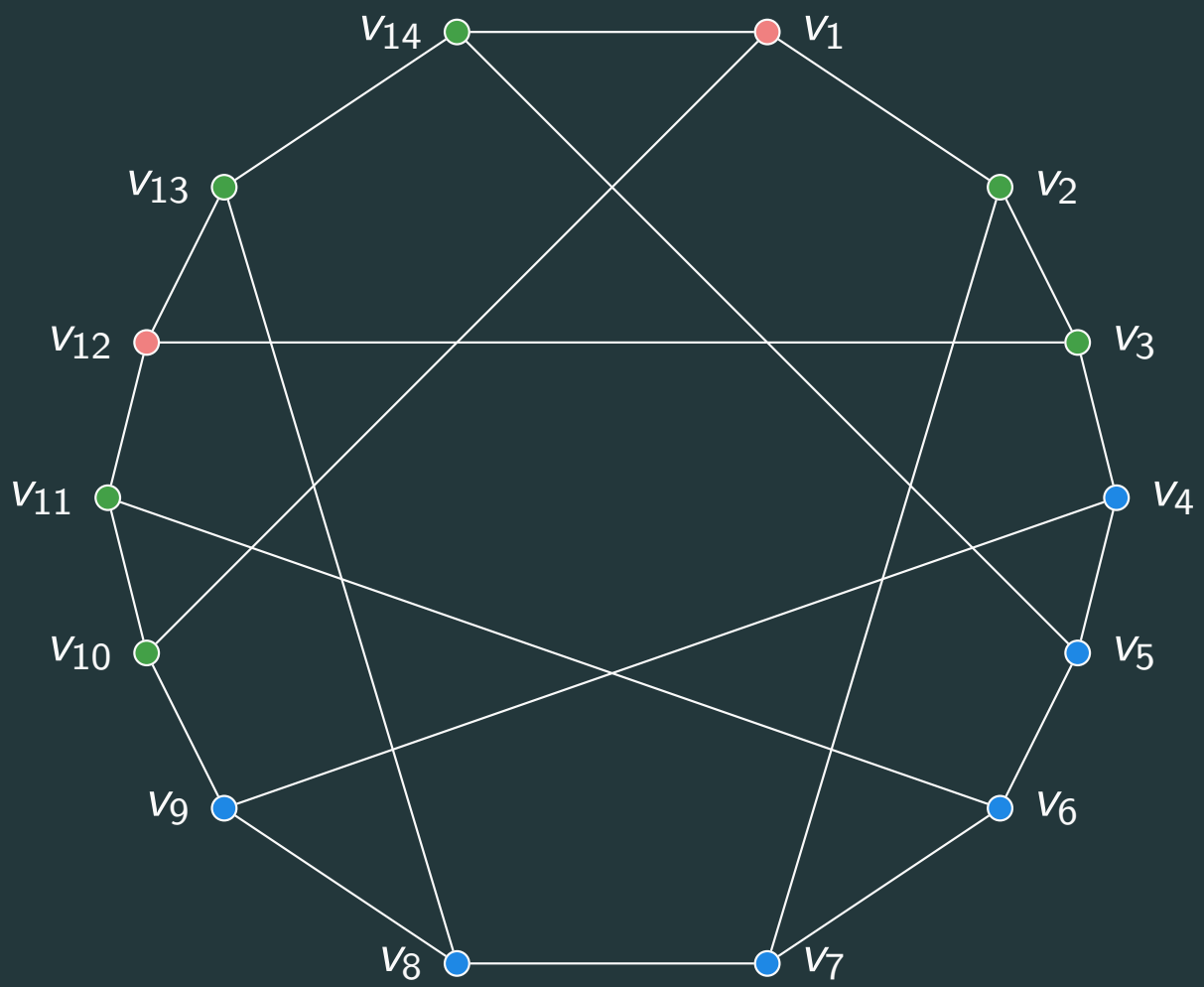


Figure 9: Haewood Graph

Literature

Power Domination Number: γ_P

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Theorem

[?] For any graph G, $1 \leq \gamma_P(G) \leq \gamma(G)$.

Power Domination in Paths: $\gamma_P(P_n) = 1$



Figure 10: P_7

Power Domination in Cycles: $\gamma_P(C_n) = 1$

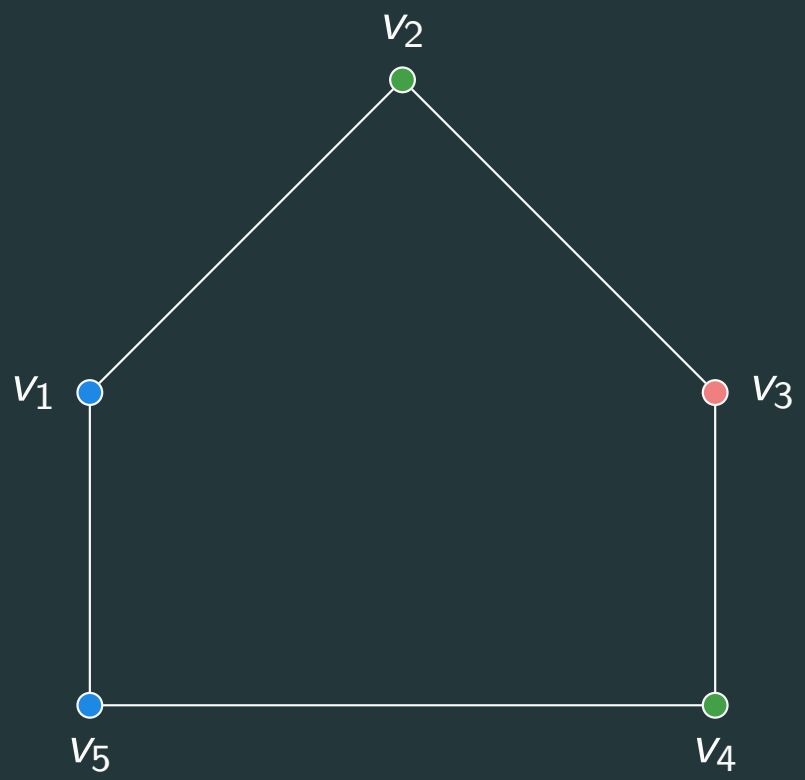


Figure 11: C_5

Other Graphs: $\gamma_P(K_n) = \gamma_P(W_n) = \gamma_P(K_{1,n}) = 1$

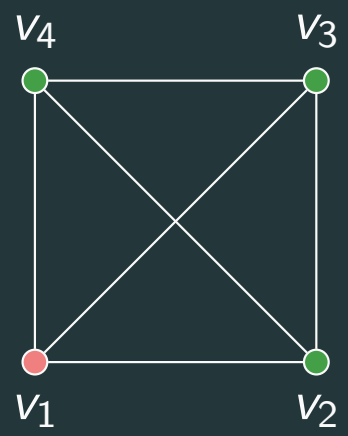


Figure 12: K_4

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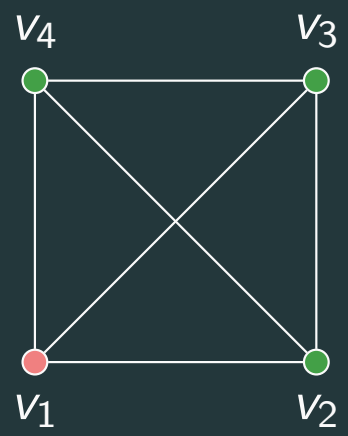


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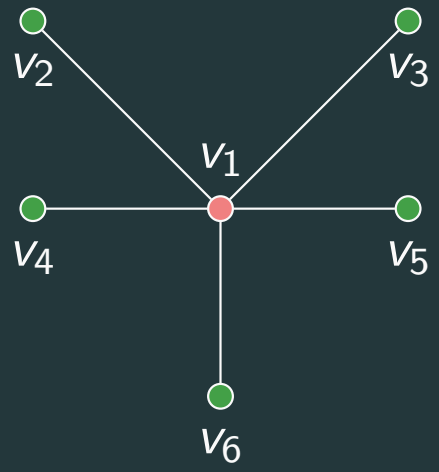


Figure 13: $K_{1,5}$

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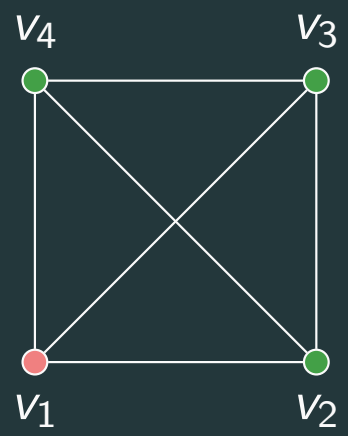


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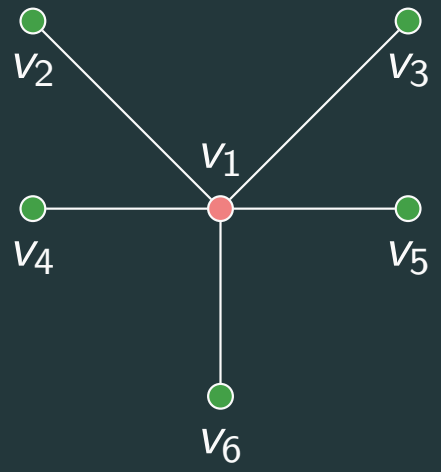


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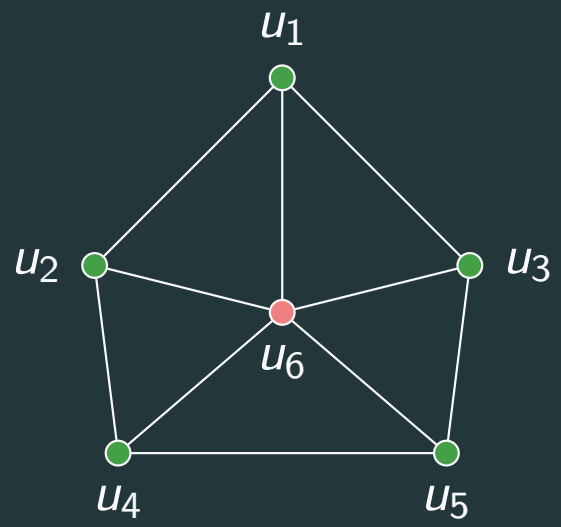


Figure 14: W_6

Cartesian Products

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  - $x_1 = x_2$ and $y_1y_2 \in E(H)$
  - $y_1 = y_2$ and $x_1x_2 \in E(G)$

Cartesian Product Example



Figure 15: P_4

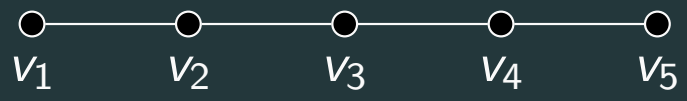


Figure 16: P_5

Cartesian Product Example

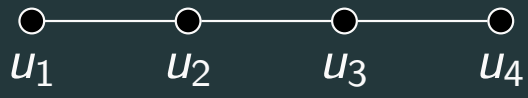


Figure 15: P_4



Figure 16: P_5

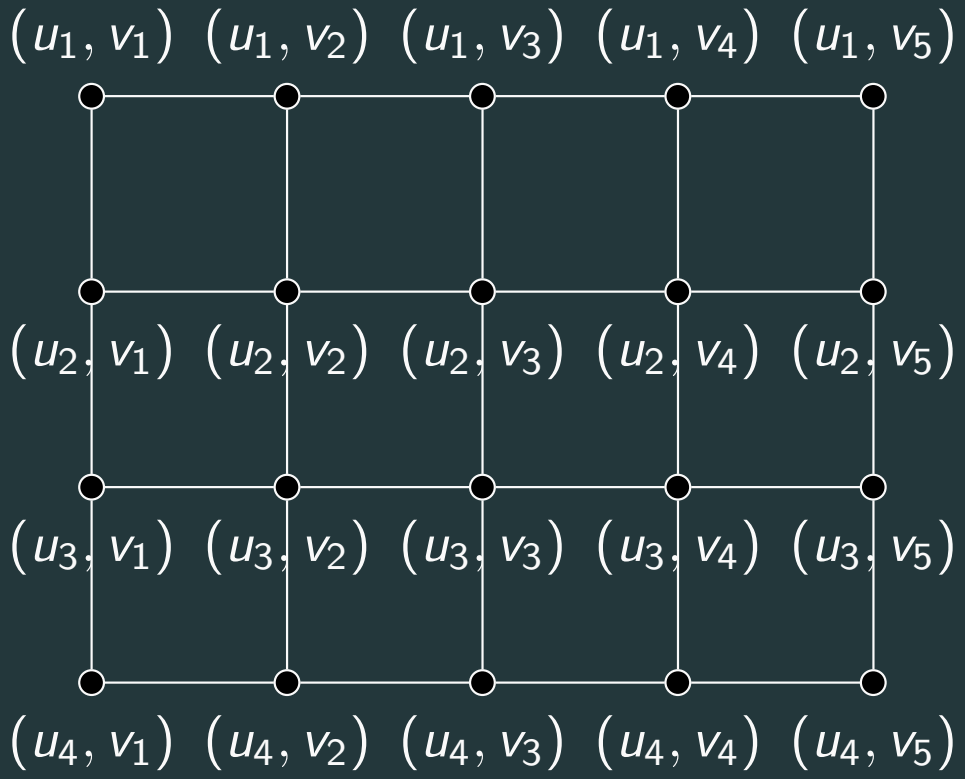


Figure 17: $P_4 \square P_5$

Global Upper Bound for Cartesian Products

Theorem

For any graphs G and H,

$$\gamma_P(G\square H) \leq \min\{\gamma_P(G)|V(H)|, \gamma_P(H)|V(G)|, \gamma(G)(|V(H)|-1), \gamma(H)(|V(G)|-1)\}.$$

Proof: $(\gamma_P(P_4))|V(K_{1,3})|$

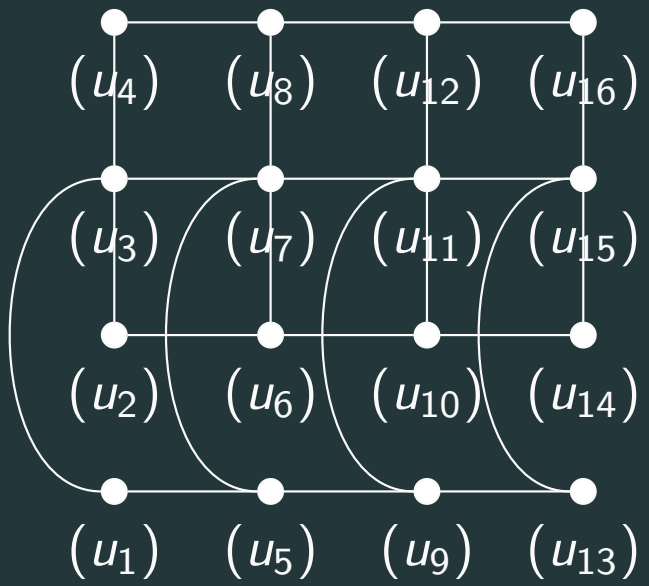


Figure 18: $P_4 \square K_{1,3}$

Proof: $(\gamma_P(P_4))|V(K_{1,3})|$

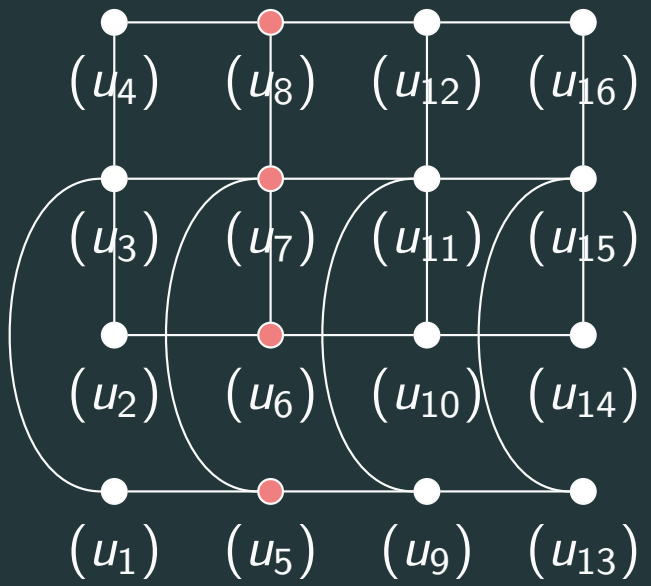


Figure 19: $P_4 \square K_{1,3}$

Proof: $(\gamma_P(P_4))|V(K_{1,3})|$

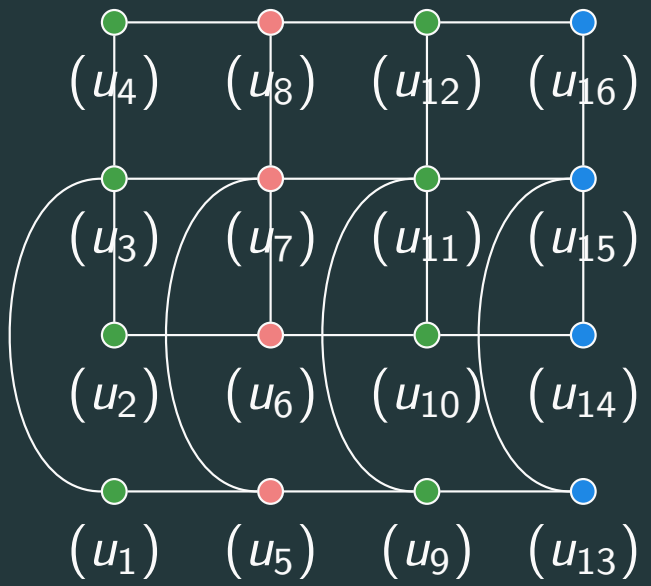


Figure 20: $P_4 \square K_{1,3}$

Proof: $(\gamma_P(K_{1,3}))|V(P_4)|$

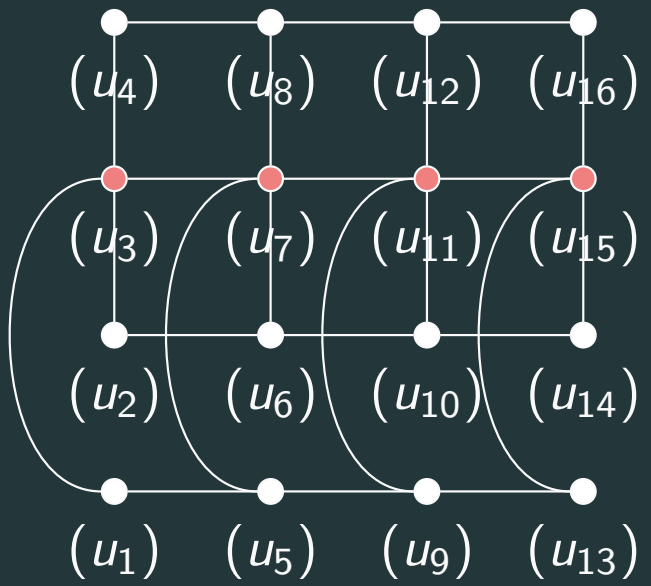


Figure 21: $P_4 \square K_{1,3}$

Proof: $(\gamma_P(K_{1,3}))|V(P_4)|$

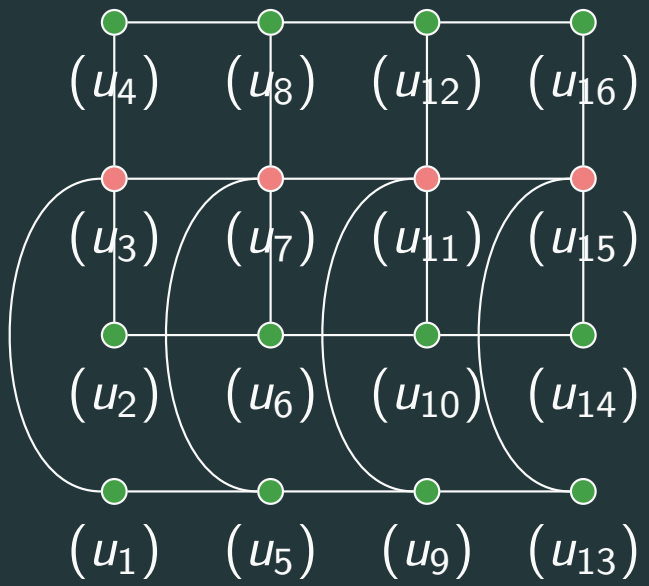


Figure 22: $P_4 \square K_{1,3}$

Proof: $(\gamma(P_4))|V(K_{1,3})| - 1$

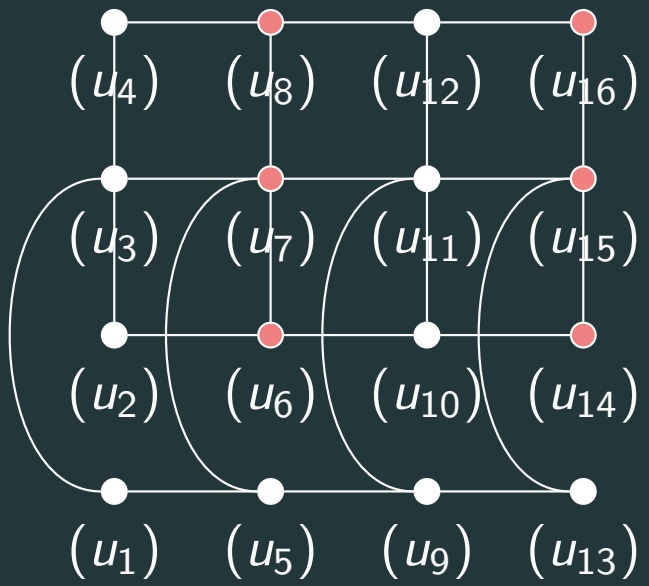


Figure 23: $P_4 \square K_{1,3}$

Proof: $(\gamma(P_4))|V(K_{1,3})| - 1$

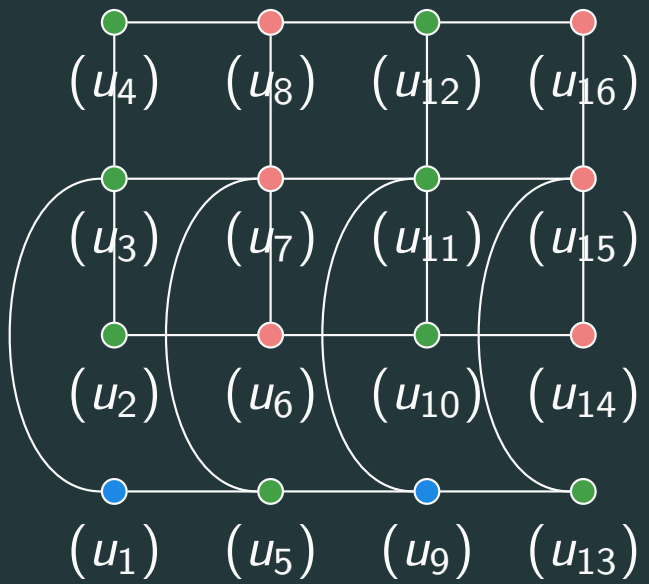


Figure 24: $P_4 \square K_{1,3}$

Proof: $(\gamma_P(K_{1,3}))|V(P_4)|$

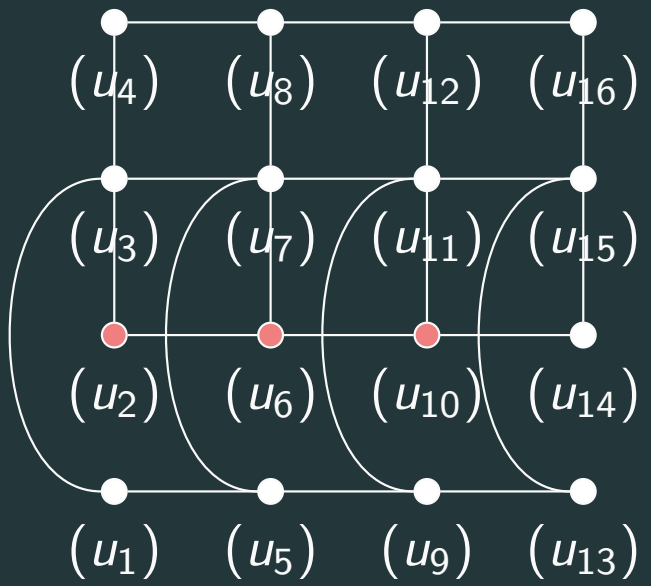


Figure 25: $P_4 \square K_{1,3}$

Proof: $(\gamma_P(K_{1,3}))|V(P_4)|$

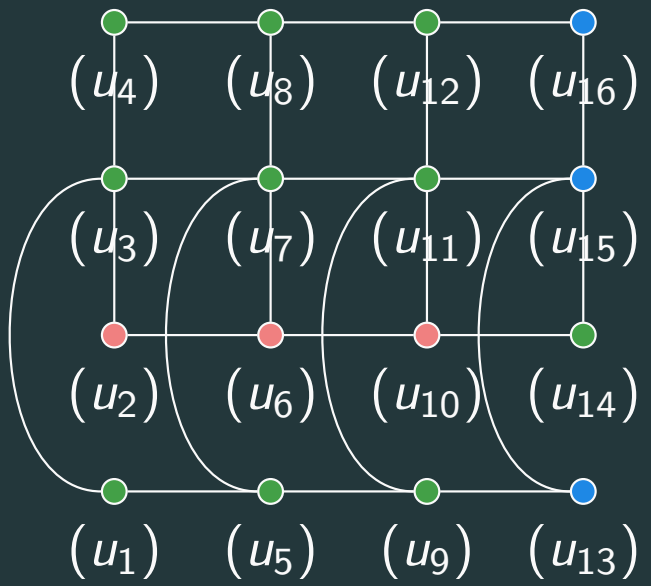


Figure 26: $P_4 \square K_{1,3}$

Other Results

Theorem

For $m \geq n \geq 3, \gamma_p(K_n \square K_{1,m}) = n - 1.$

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For $n \geq 4$, $\gamma_P(W_n \square W_n) = 3$.

Product of Three Graphs

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Problem
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Figure 27: $P_3 \Box P_3 \Box P_3$

Product of Three Graphs

Theorem

For any $n \geq 3$, $\gamma_P(C_n \Box C_n \Box C_n) \leq n^2$.

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For $n \geq 1$, $\gamma_P(K_{1,n} \Box K_{1,n} \Box K_{1,n}) \leq n(n-1)$.

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For $n \geq 3$, $\gamma_P(W_n \Box W_n \Box W_n) \leq 3n$.

Cubic Graphs

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Cubic graphs are graphs all of whose vertices have degree 3 and therefore are 3-regular graphs. The order of a cubic graph is always even. A cubic graph of 8 vertices is shown in Figure 28.

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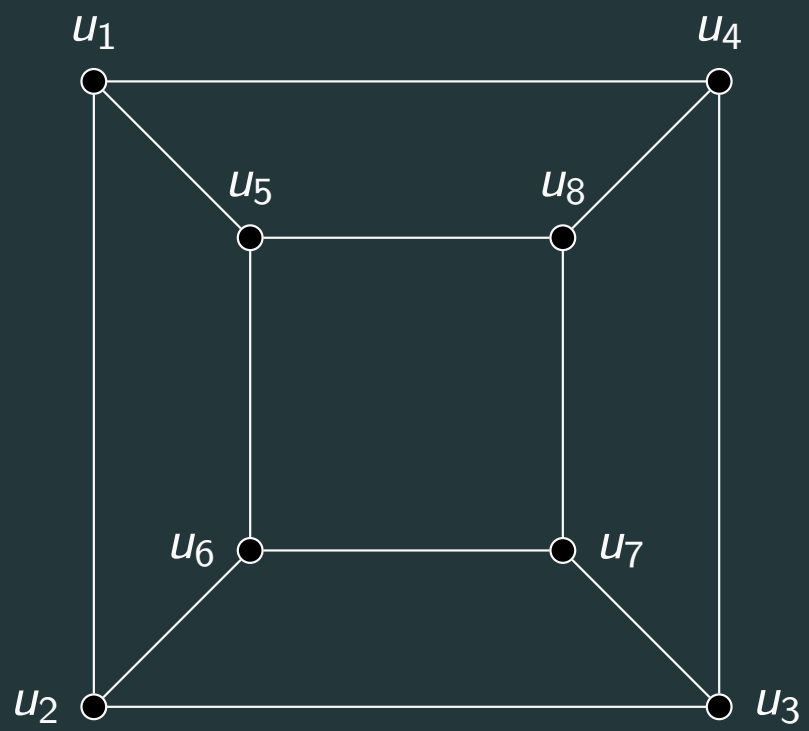


Figure 28: Cubic Graph on 8 vertices

Global Upper Bound on Cubic Graphs

Theorem

Suppose G is a 2-connected cubic multigraph of order n. For any vertex $x \in V(G)$, there exists a power dominating set of cardinality at most $n/3$ containing x.

Cubic Graphs with $\gamma_P = 1$

The girth of a graph G, denoted by $g(G)$, is the length of the shortest graph cycle (if any) in G.

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The girth of a graph G, denoted by $g(G)$, is the length of the shortest graph cycle (if any) in G.

Lemma

A cubic graph G with $g(G) > 3$ will have $\gamma_P(G) > 1$.

It follows that if G is cubic with $\gamma_P(G) = 1$, then G must contain a triangle.

Franklin Graph

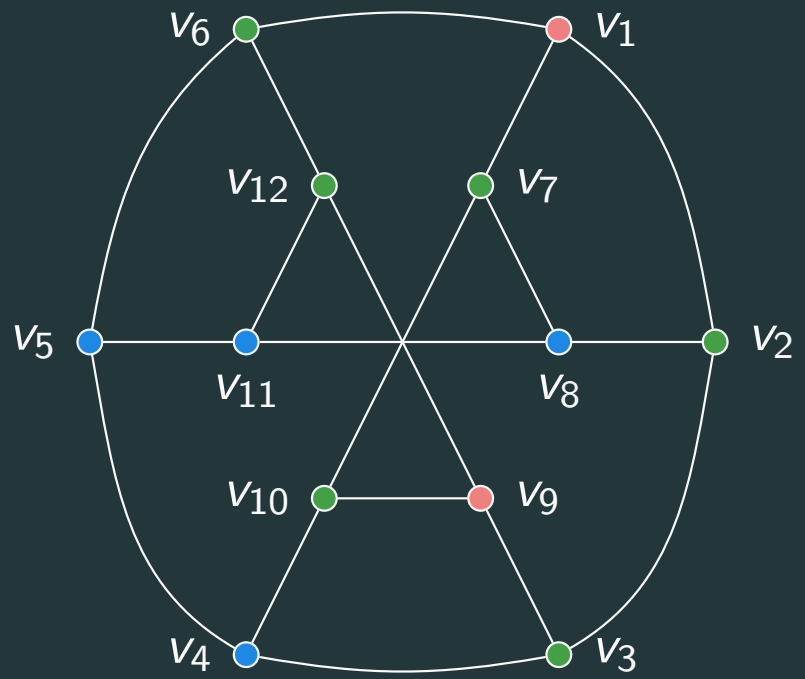


Figure 29: Power dominating set of the Franklin Graph

Möbius–Kantor graph

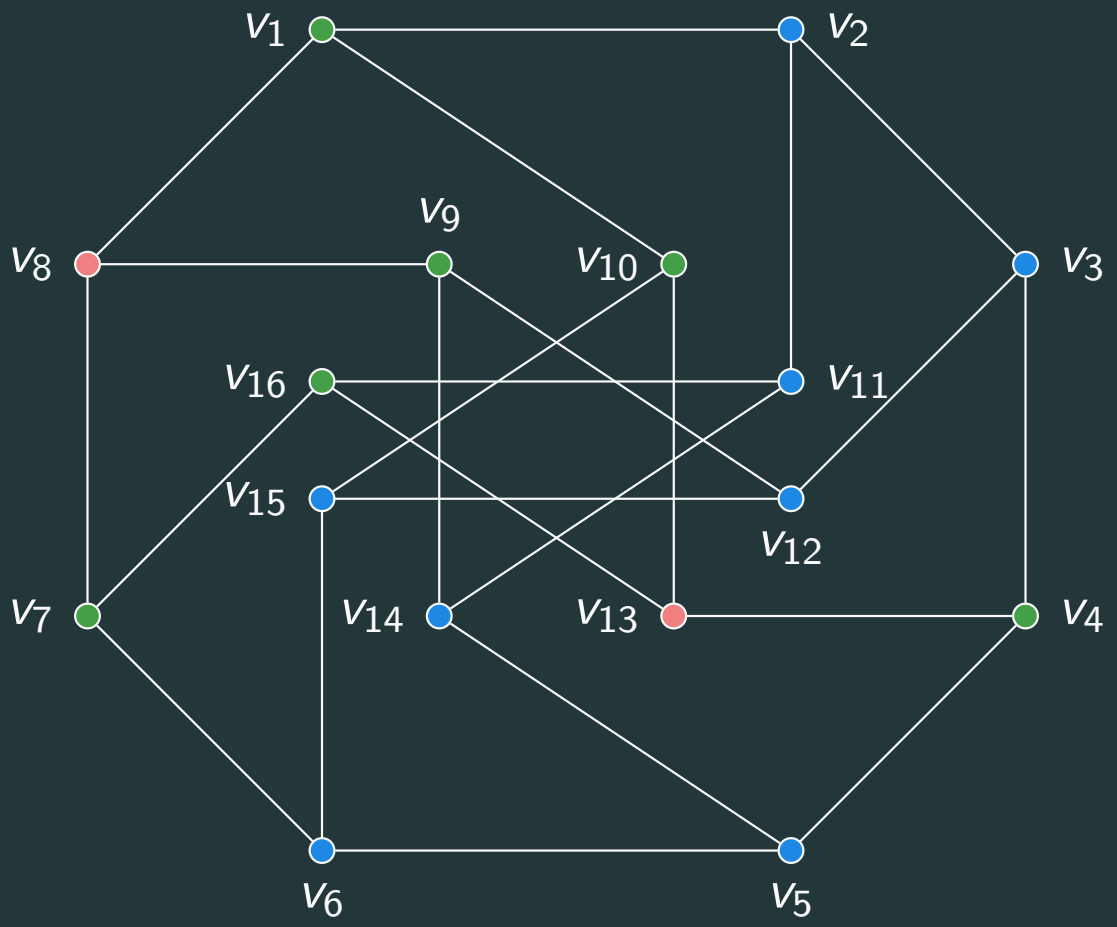


Figure 30: Möbius–Kantor Graph

Zero-symmetric graph

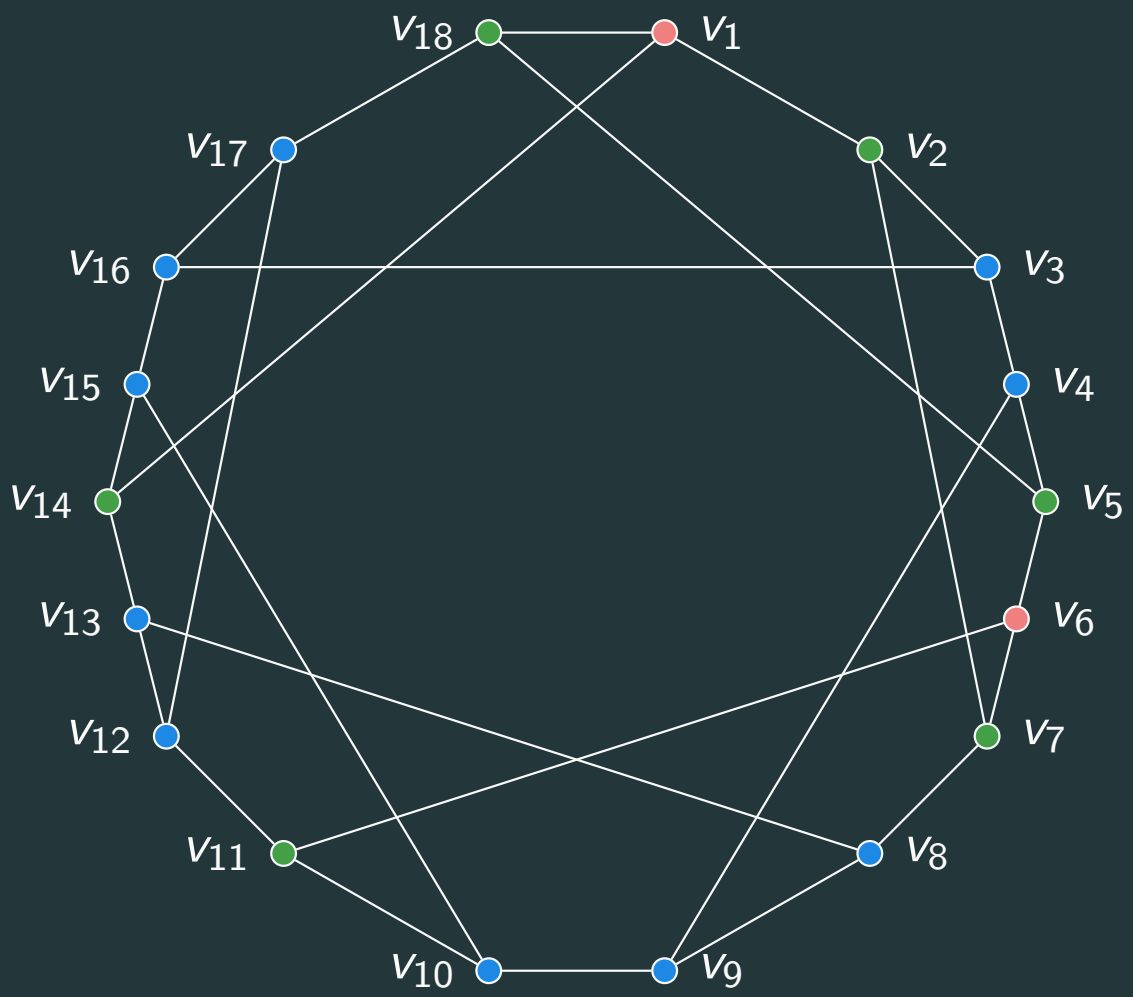


Figure 31: Zero-symmetric Graph

LCF Notation

A Hamiltonian cycle is a graph cycle through a graph that visits each vertex exactly once.

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LCF notation represented the adjacency of vertex in a cubic graph with a Hamiltonian cycle.

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LCF notation represented the adjacency of vertex in a cubic graph with a Hamiltonian cycle.

For example, LCF notation of $[2, 6, -2]^4$ gives the following adjacencies:

Vertex	**Adjacenies**
v_1	v_3
v_2	v_8
v_3	v_1

LCF Notation

All of the graphs in the table below are cubic, symmetric, triangle-free, bipartite and Hamiltonian.

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Graph	Number of Vertices	LCF Notation
Franklin Graph	12	$[5, -5]^6$
Haewood Graph	14	$[5, -5]^7$
Möbius–Kantor Graph	16	$[5, -5]^8$
Zero-symmetric Graph	18	$[5, -5]^9$

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All of the graphs in the table below are cubic, symmetric, triangle-free, bipartite and Hamiltonian.

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Zero-symmetric Graph	18	$[5, -5]^9$

These graphs have LCF notation of $[5, -5]^n$ for $n < 10$ and $\gamma_P(G) = 2$.

Conjecture
For a graph G with LCF notation $[5, -5]^n$ for $n \geq 10$, $\gamma_P(G) = 2$.

Cubic Circulant Graphs

A circulant graph denoted by $Ci_n(\ell)$ is a graph of order $n \geq 3$ with list $\ell \subseteq [n]$ such that:

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- $V(Ci_n(\ell)) = \{u_1, \ldots, u_n\}$
- Each vertex u_i is adjacent to:
  - u_{i+j} and u_{i-j} for each $j \in \ell$

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For a circulant graph to be cubic, each vertex in $Ci_n(\ell)$ must be adjacent to 3 vertices and n must be even. So, $\ell = \{k, \frac{n}{2}\}$ where $1 \leq k < \frac{n}{2}$

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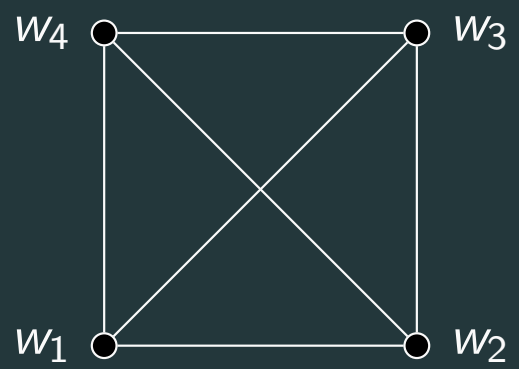


Figure 32: $K_4 = Ci_4(1, 2)$

Circulant Graphs

We studied modular values of n and k to determine the $\gamma_P(Ci_n\{k, \frac{n}{2}\})$ for $n > 4$.

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k	n	$\gamma_P(Ci_n\{k, \frac{n}{2}\})$
1	$n \equiv 0 \pmod 2$	2
$k \not\equiv 0 \pmod 2$	$n \not\equiv 0 \pmod k$	2
$k \equiv 0 \pmod 2$	$n \equiv 0 \pmod 4$	4
$k \equiv 0 \pmod 2$	$n \not\equiv 0 \pmod 4$	2
$k > 1$	$n \equiv 0 \pmod k$	2

Mycielski Graphs

Mycielski Construction [?]

- Suppose G is a graph with $V(G) = \{v_1, \ldots, v_n\}$

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- The Mycielskian of G is the graph with
  1. Vertex set $V(G) \cup V' \cup \{w\}$, where $V' = \{u_i \mid v_i \in V(G)\}$
  2. Edge set $E(G) \cup \{u_i v_j \mid v_i v_j \in E(G)\} \cup \{w u_j \mid u_j \in V'\}$

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  2. Edge set $E(G) \cup \{u_i v_j \mid v_i v_j \in E(G)\} \cup \{w u_j \mid u_j \in V'\}$
- The Mycielskian of G is denoted by $\mu(G)$

$\mu(P_4)$

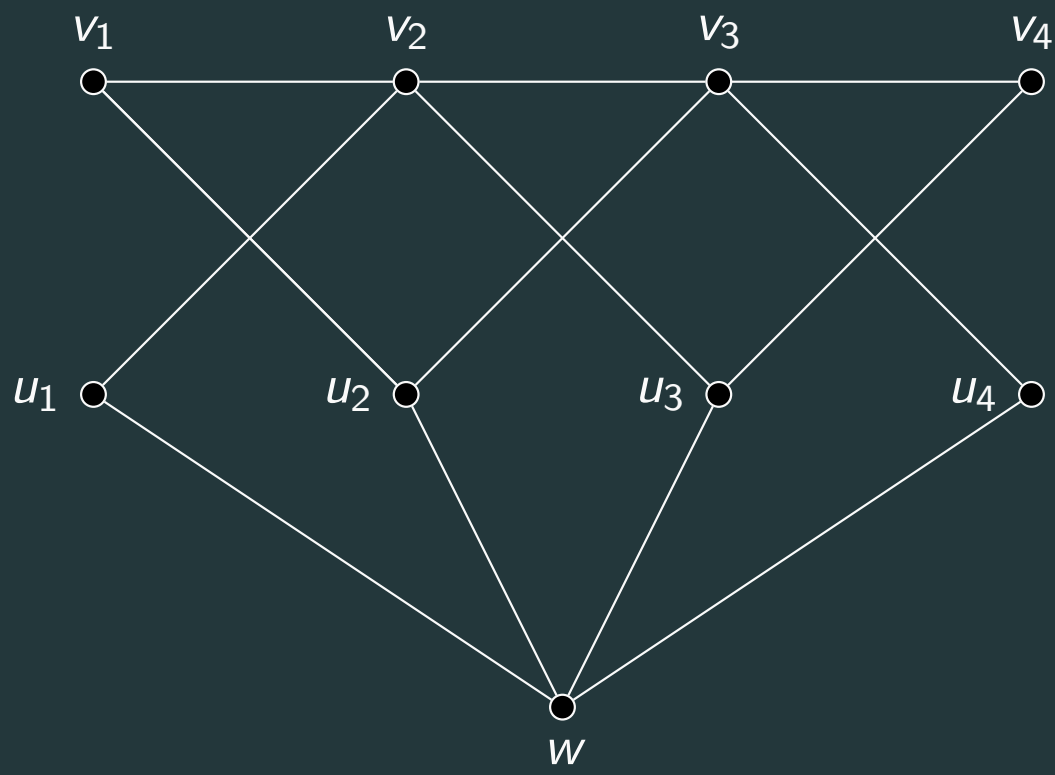


Figure 33: $\mu(P_4)$

$\gamma_p(\mu(P_n))$

Theorem

$$\text{For } n \geq 2,\ \gamma_P(\mu(P_n)) = \begin{cases} 1 & \text{if } n \equiv 0 \pmod 2 \\ 2 & \text{if } n \equiv 1 \pmod 2 \end{cases}$$

Other Mycielski Graphs

Theorem

For $n \geq 4$, $\gamma_P(\mu(C_n)) = 2$.

Other Mycielski Graphs

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For $n \geq 4$, $\gamma_P(\mu(C_n)) = 2$.

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For $n \geq 2, \gamma_P(\mu(K_n)) = 1$.

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Theorem

For $n \geq 2, \gamma_P(\mu(K_n)) = 1$.

Theorem

For $n \geq 1$, $\gamma_P(\mu(K_{1,n})) = 1$.

Other Mycielski Graphs

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For $n \geq 4$, $\gamma_P(\mu(C_n)) = 2$.

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For $n \geq 2, \gamma_P(\mu(K_n)) = 1$.

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For $n \geq 1$, $\gamma_P(\mu(K_{1,n})) = 1$.

Theorem

For $n \geq 4$, $\gamma_P(\mu(W_n)) = 1$.

Conclusion

Open Questions:

1. Suppose G is a cubic graph with LCF notation $[k, -k]^n$ and H is a cubic graph with LCF notation $[k, -k]^m$. Is it true that $\gamma_P(G) = \gamma_P(H)$?

Open Questions:

1. Suppose G is a cubic graph with LCF notation $[k, -k]^n$ and H is a cubic graph with LCF notation $[k, -k]^m$. Is it true that $\gamma_P(G) = \gamma_P(H)$?
2. Are there classes of graphs, for which the $\gamma_P(G) \leq sp(T)$ where T is a spanning tree of G?

Thank You

Questions?

Backup slides

Examples and Counterexamples

References i